Análisis Matemático, Complejidad Asintótica y Graficación

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Abstract

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Keywords: keyword1, Keyword2, Keyword3, Keyword4

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2 Results

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3 Equations

Listing 1 Código A

```
public void funcionA(int n) {
    int total = 0;
    for (int i = 0; i < n; i++) {
        total += i;
    }
    System.out.println(total);
}
```

Listing 2 Código B

```
public void funcionB(int n) {
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < n; j++) {
            System.out.println("Procesando coordenadas:" + i + "," + j);
        }
    }
}
```

Listing 3 Código C

```
public void funcionC(int n) {
    int actual = n;
    while (actual > 1) {
        System.out.println("Valor:" + actual);
        actual = actual / 2;
    }
}
```

1. Conteo de operaciones elementales.

En este ejercicio, las operaciones elementales son las siguientes:

- `int actual = n`: $C_1 \times 1$
- `while (actual > 1)`: n
- `System.out.println("Valor:" + actual)`: $C_2 \times 1$
- `actual = actual / 2`: $C_3 \to \log n$

1. Cálculo de la complejidad asintótica.

Función C:

$$T(n) = C1 + n(C2 + \log n) = C1 + \log n = O(n)$$

Función C:

$$T(n) = C1 + \log n$$

Función C:

$$T(n) = \log n$$

$$O(\log n)$$

4 Graphs

Tables can be inserted via the normal table and tabular environment. To put

Sample body text. Sample body text. Sample body text. Sample body text. Sample body text (refer Figure **??**). Sample body text. Sample body text. Sample body text (refer Table **??**).

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7 Conclusion

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Appendix A Section title of first appendix

An appendix contains supplementary information that is not an essential part of the text itself but which may be helpful in providing a more comprehensive understanding of the research problem or it is information that is too cumbersome to be included in the body of the paper.

References

[1] Campbell, S.L., Gear, C.W.: The index of general nonlinear DAES. Numer. Math. **72**(2), 173–196 (1995)